HLM Assignment 3
Due May 17 at 10 p.m.

Part A. Ordered Logit Models

We are revisiting our inquiry into the effect of Catholic schools on educational outcomes, this time considering educational attainment with a bit more sophistication. The data file is called nels_HW3.mdm. The outcome variable is DEGREE, coded 0= BA gradate, 1= high school graduate, 2= less than high school. Note that higher education attainment is coded as lower values. This is somewhat unconventional but done to make the interpretation of coefficients more intuitive, as they were in the example presented in class.

1. Run an ordered logistic HGLM with degree being predicted by race, gender, and base year SES. Leave each of these variables uncentered and with fixed slopes. Interpret the intercept, the coefficient for Black, the coefficient for SES, and δ_2.
2. What are the log odds of a black student with SES=1 having obtained a college degree? What about a high school degree (this includes college graduates)? Explain your calculation.
3. Is the black-white gap in degree attainment significantly different from the Hispanic-white gap? Use a Wald test and report your results.
4. Add Catholic at level 2 for each level-1 term. Run the model and discuss what you've learned in relation to our research question.

Part B. Cross-classified Models

We consider ecological disadvantage, this time measured by "neighborhood social deprivation" in Scotland (see the "attain" data), and its effects on educational outcomes. The outcome variable is ATTAIN, which you can refer to as a composite measure of achievement.

1. Write down a model that controls for prior reading achievement (P7READ), father's occupation and employment status, and gender. Interpret the intercept, coefficients, and random effects.
2. Use HLM to estimate the parameters of this model. Do you find any association between neighborhood social deprivation and achievement, controlling for relevant covariates? Provide statistical evidence to support your conclusion.
3. Do you find any evidence that the effect of neighborhood deprivation varies by school attended?
4. Does the effect of neighborhood deprivation change the effects of any of your child-level covariates? Create a model graph to show this interaction effect, and explain how the shape of the lines depicted in the graph reflect the estimates of parameters in your model. (If you did not find any significant interaction effects, select either continuous level-1 variable for your Z-focus).

You may use a single model to address all of these questions. Explain the statistical evidence for your answers.